

TOPIC 0

Mathematical background for AI

Explanatory study notes

Theory, worked examples and revision questions

What you should be able to do

Use conditioning, marginalization and Bayes' rule; calculate expectations and Gaussian predictions; connect entropy, KL divergence and cross-entropy to learning; understand how constraints become regularization.

Based on the teacher PDFs supplied in **Lecture Slides**. The notes explain and reorganize that material. Worked calculations are study examples derived from the slide concepts, unless explicitly identified as a slide exercise. Clarifications of ambiguous or incorrect expressions are marked in context. References use **PDF page numbers**, counted from 1, rather than printed slide numbers.

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1 Why these tools appear throughout the course

Teacher slides: 0.1. Math background, pp. 1–11.

Machine learning models must handle incomplete information and noisy measurements. Probability represents uncertainty; information theory measures how well a model describes observations; optimization chooses parameters. These are connected: a probability model defines a likelihood, the negative log-likelihood becomes a loss, and an optimizer minimizes it.

Use $p(x)$ for a probability mass or density, depending on whether X is discrete or continuous. Uppercase X denotes a random variable; lowercase x is one realized value. A vector is written in bold when helpful. E is expectation, Var is variance, and Σ is a covariance matrix. All logarithms in these notes are natural unless a base is stated.

A chain of ideas to remember

Probability model $\rightarrow$ likelihood $\rightarrow$ negative log-likelihood $\rightarrow$ loss $\rightarrow$ optimization. A prior adds assumptions about plausible parameters and can produce a regularization term.

2 Conditioning, joint probability and marginalization

Teacher slides: 0.1. Math background, pp. 1–2.

2.1 Conditioning changes the information available

For an event B with positive probability,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(A \cap B) = P(A | B)P(B).$$

Conditioning restricts attention to outcomes where B occurred. It is directional: $P(A | B)$ and $P(B | A)$ usually differ. Independence means $P(A \cap B) = P(A)P(B)$, so conditioning on B does not change the probability of A .

For a sequence, the chain rule gives

$$p(x_1, \dots, x_n) = p(x_1) \prod_{i=2}^n p(x_i | x_1, \dots, x_{i-1}).$$

The chain rule always holds; simplifying each conditional to $p(x_i)$ requires independence. This distinction will matter for RL trajectories, where successive states are dependent.

2.2 Worked slide exercise: drawing balls

The slides give an urn containing three red balls and one blue ball. Draw without replacement. The chance of two reds is

$$P(R_1, R_2) = P(R_1)P(R_2 | R_1) = \frac{3}{4} \frac{2}{3} = \frac{1}{2}.$$

The second probability is $2/3$, because one red has already been removed. Using $3/4$ twice would incorrectly assume independence.

To find the chance that the second ball is red, sum over the possible first draws:

$$\begin{aligned} P(R_2) &= P(R_2 | R_1)P(R_1) + P(R_2 | B_1)P(B_1) \\ &= \frac{2}{3} \frac{3}{4} + 1 \frac{1}{4} = \frac{3}{4}. \end{aligned}$$

This is marginalization: remove a variable by summing over all its possible values. For a partition $B_1, \dots, B_K$,

$$P(A) = \sum_j P(A | B_j)P(B_j).$$

For densities, replace the sum with an integral: $p(x) = \int p(x, y) dy$.

3 Bayes' rule: reverse a conditional probability

Teacher slides: 0.1. Math background, p. 2.

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

In model language, the posterior is proportional to likelihood times prior:

$$p(h | D) = \frac{p(D | h)p(h)}{\sum_j p(D | h_j)p(h_j)}.$$

The **prior** describes uncertainty before these data; the **likelihood** describes how the assumed model generates these data; the **posterior** incorporates both. The denominator, called evidence, normalizes the result. Continuous parameters require integration instead of summation.

3.1 Worked model-identification example

Choose with equal probability between a fair coin F and a biased coin B with $P(H | B) = 0.9$. Observe one head. Then

$$P(B | H) = \frac{0.9(0.5)}{0.9(0.5) + 0.5(0.5)} = \frac{9}{14} \approx 0.643.$$

One head provides evidence for the biased coin but does not identify it with certainty. After two independent heads conditional on the chosen coin,

$$P(B | HH) = \frac{0.9^2}{0.9^2 + 0.5^2} \approx 0.764.$$

The two draws are conditionally independent given the coin. Their marginal dependence comes from sharing the same unknown coin.

4 Random variables, expectation and variance

Teacher slides: 0.1. Math background, pp. 2–4.

4.1 Mass, density and cumulative probability

A discrete variable has probabilities $p(x) \geq 0$ that sum to one. A continuous variable has a density $p(x) \geq 0$ whose integral is one. For a continuous variable,

$$P(a \leq X \leq b) = \int_a^b p(x) dx, \quad F(x) = P(X \leq x) = \int_{-\infty}^x p(t) dt.$$

At differentiable points, $F'(x) = p(x)$. A density value is not a point probability: for a continuous variable $P(X = x) = 0$, and a density can exceed one on a narrow interval.

4.2 Expectation is a probability-weighted average

$$E[g(X)] = \sum_x g(x)p(x) \quad \text{or} \quad E[g(X)] = \int g(x)p(x) dx.$$

Linearity gives $E[aX + bY] = aE[X] + bE[Y]$ without needing independence. In contrast, $E[XY] = E[X]E[Y]$ generally needs an assumption such as independence.

$$\mu = E[X], \quad \text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2.$$

For two variables,

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

Independence removes the covariance term. Zero covariance alone does not generally imply independence.

4.3 Bernoulli and indicator variables

For $X \in \{0, 1\}$ with success probability θ ,

$$p(x) = \theta^x(1 - \theta)^{1-x}, \quad E[X] = \theta, \quad \text{Var}(X) = \theta(1 - \theta).$$

An indicator I_A is one on event A and zero elsewhere, so $E[I_A] = P(A)$. Logistic classification will use Bernoulli outputs, while nearest-neighbour voting can be written as a sum of class indicators.

5 Gaussian distributions and linear transformations

Teacher slides: 0.1. Math background, pp. 4–6.

5.1 Univariate normal

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu)^2}{2\sigma^2}\right].$$

The second argument is the *variance*, not the standard deviation. Standardization gives $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$. Conversely, $X = \mu + \sigma Z$ has mean μ and variance σ^2 .

Independent Gaussian variables have a Gaussian sum. Its mean is the sum of means and its variance is the sum of variances. More generally, a linear transformation of a jointly Gaussian vector is Gaussian.

5.2 Multivariate normal

For $\mathbf{x} \in \mathbb{R}^d$ and positive-definite covariance Σ ,

$$p(\mathbf{x}) = \frac{\exp[-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})]}{(2\pi)^{d/2} |\Sigma|^{1/2}}.$$

The quadratic form measures distance while accounting for scale and correlations. The log-density is

$$\log p(\mathbf{x}) = -\frac{d}{2} \log(2\pi) - \frac{1}{2} \log |\Sigma| - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu}).$$

For $Y = A\mathbf{X} + \mathbf{b}$, mean and covariance become $A\boldsymbol{\mu} + \mathbf{b}$ and $A\Sigma A^T$. If $\mathbf{Z} \sim \mathcal{N}(0, I)$ and $LL^T = \Sigma$, then $\boldsymbol{\mu} + L\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$. A Cholesky factor L is a valid sampling factor, even though it is generally not the symmetric matrix square root.

5.3 Conditioning a joint Gaussian

Partition the mean and covariance into blocks for X_a, X_b . Then

$$\begin{aligned} X_b \mid X_a = x_a &\sim \mathcal{N}(\mu_{b|a}, \Sigma_{b|a}), \\ \mu_{b|a} &= \mu_b + \Sigma_{ba} \Sigma_{aa}^{-1} (x_a - \mu_a), \\ \Sigma_{b|a} &= \Sigma_{bb} - \Sigma_{ba} \Sigma_{aa}^{-1} \Sigma_{ab}. \end{aligned}$$

The conditional mean adjusts according to correlation; the covariance decreases because an observation adds information. This identity is the basis of Gaussian-process prediction.

Clarifications of the Gaussian summary on pp. 5–6

The conditional covariance uses a **minus** sign. A weighted mixture of Gaussian densities is generally not Gaussian. Gaussian marginals alone do not guarantee a jointly Gaussian distribution. A linear combination of correlated Gaussian blocks must include cross-covariance terms. Also, $\boldsymbol{\mu} + \Sigma^{1/2}\mathbf{Z}$ has covariance Σ , rather than Σ^2 . These distinctions prevent errors when using the compact slide summary.

5.4 Worked conditioning example

Let (X, Y) have zero means, unit variances and covariance 0.8. For $X = 1$,

$$Y \mid X = 1 \sim \mathcal{N}(0.8, 1 - 0.8^2) = \mathcal{N}(0.8, 0.36).$$

The conditional standard deviation is 0.6. The prediction is uncertain even after X is known, because correlation is not perfect.

6 Entropy, cross-entropy and KL divergence

Teacher slides: 0.1. Math background, pp. 6–9.

6.1 Surprise and entropy

The surprise of an outcome with probability $p(x)$ is $-\log p(x)$. Entropy is expected surprise:

$$H(p) = -\sum_x p(x) \log p(x).$$

With $\log_2$, units are bits; with natural logs, they are **nats** (the handout calls them “nits”). Use the convention $0 \log 0 = 0$.

For a Bernoulli variable,

$$H(\theta) = -\theta \log \theta - (1 - \theta) \log(1 - \theta).$$

A fair coin has entropy $\log 2 \approx 0.693$ nats, or one bit. A deterministic coin has entropy zero. Differential entropy for continuous variables replaces the sum by an integral, but can be negative and depends on the measurement scale.

6.2 Cross-entropy evaluates a model against data

If data follow p and a model assigns probabilities q ,

$$H(p, q) = -\sum_x p(x) \log q(x), \quad \text{KL}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}.$$

Therefore,

$$\boxed{H(p, q) = H(p) + \text{KL}(p\|q)}.$$

KL is nonnegative and is zero when distributions agree, apart from null sets. It is not symmetric and does not satisfy the requirements of a distance metric. If q assigns zero probability to an event with positive p , the divergence is infinite.

6.3 Worked slide exercise: two coins

For $P = (0.5, 0.5)$ and $Q = (0.9, 0.1)$,

$$\text{KL}(P\|Q) = 0.5 \log(0.5/0.9) + 0.5 \log(0.5/0.1) \approx 0.511,$$

$$\text{KL}(Q\|P) = 0.9 \log(0.9/0.5) + 0.1 \log(0.1/0.5) \approx 0.368.$$

The larger first value reflects the cost of using a model that treats tails as rare when tails actually occur half the time.

6.4 Information gain: a careful distinction

The handout links posterior-versus-prior KL to entropy reduction. For a *particular* observed y , define the realized Bayesian information gain as $\text{KL}(p(X | y)\|p(X))$. It is not generally equal to $H(X) - H(X | Y = y)$.

The equality holds *after averaging over observations*:

$$I(X; Y) = \mathbb{E}_Y[\text{KL}(p(X | Y)\|p(X))] = H(X) - H(X | Y).$$

This expected entropy reduction is the information-gain idea used in decision-tree splitting. A particular surprising observation can increase posterior entropy; the expected reduction is nonnegative.

7 Likelihood, estimation and optimization

Teacher slides: 0.1. Math background, pp. 8–11.

7.1 Maximum likelihood and cross-entropy

For independent observations conditional on parameters w ,

$$L(w) = \prod_{i=1}^n p(x_i | w), \quad \hat{w}_{\text{ML}} = \arg \max_w \sum_i \log p(x_i | w).$$

Taking logs preserves the maximizer and turns products into sums. Dividing by n does not alter the maximizer either. The negative average log-likelihood is the empirical cross-entropy between the observed distribution and the model. Its population counterpart differs from KL only by the entropy of the true distribution, which is independent of w .

Maximum a posteriori estimation instead maximizes $\log L(w) + \log p(w)$. Full Bayesian inference retains $p(w | D)$ rather than only its maximizing point.

A qualification to the MLE statement on p. 9

MLE need not be unbiased in a finite sample. Consistency and asymptotic normality require regularity and correct-model conditions. With per-observation Fisher information $I(w_*)$, the usual statement is $\sqrt{n}(\hat{w} - w_*) \Rightarrow \mathcal{N}(0, I(w_*)^{-1})$. Identifiability and sample size matter.

7.2 A norm constraint becomes ridge regularization

Start from

$$\min_w f(w) \quad \text{subject to} \quad \|w\|_2^2 \leq C.$$

The Lagrangian is $\mathcal{L}(w, \lambda) = f(w) + \lambda(\|w\|_2^2 - C)$ with $\lambda \geq 0$. For fixed λ , the term $-\lambda C$ does not depend on w , giving a penalized objective $f(w) + \lambda\|w\|_2^2$.

At a differentiable optimum, stationarity is $\nabla f(w) + 2\lambda w = 0$. The full constraint conditions also require feasibility and complementary slackness: $\lambda(\|w\|_2^2 - C) = 0$. An inactive constraint can have $\lambda = 0$; the tangency picture describes the active case. The correspondence between C and λ is especially useful in convex regression.

8 Revision sheet and self-test

Tool	What to ask yourself
Conditioning	What information is fixed, and is the conditioning event possible?
Marginalization	Which variable must be removed by a sum or integral?
Bayes	What are prior, likelihood and normalizing evidence?
Expectation	What outcomes are being averaged, under which distribution?
Gaussian conditioning	Are covariance blocks ordered correctly, with subtraction?

Tool	What to ask yourself
Cross-entropy	Which distribution generates data and which predicts them?
Regularization	What assumption does the penalty impose?

8.1 Questions

1. In the urn, what is $P(R_1 | R_2)$?
2. Does linearity of expectation need independent variables?
3. If $X \sim \mathcal{N}(2, 9)$, how do you standardize it?
4. Can a mixture of two Gaussians have two peaks?
5. What is cross-entropy for a correct class assigned probability 0.1?
6. Why can cross-entropy be minimized without knowing $H(p)$?
7. Does a smaller norm budget normally correspond to stronger regularization?
8. Must one particular observation decrease posterior entropy?

8.2 Answers

1. $P(R_1, R_2)/P(R_2) = (1/2)/(3/4) = 2/3$.
2. No; linearity always holds when the expectations exist.
3. $(X - 2)/3$; divide by SD, not variance.
4. Yes. A Gaussian mixture need not be Gaussian.
5. $-\log 0.1 \approx 2.303$ nats.
6. $H(p)$ is constant with respect to the model parameters.
7. Yes, in the usual convex constraint–penalty correspondence.
8. No. Entropy reduction is nonnegative in expectation.

9 Source guide

0.1. **Math background.pdf**: probability (pp. 1–6), information theory and MLE (pp. 6–9), constrained optimization (pp. 9–11). These notes derive the handout’s urn and coin exercises and clarify its compact Gaussian and information-gain summaries. No external reading is required for the explanations here.